

# POOJA CHAUDHARI

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## PROFESSIONAL SUMMARY

AI Engineer with **2+ years of experience** in building AI-based applications and data-driven solutions. Skilled in Python, machine learning, and working with LLM-based systems, including chatbots and automation workflows. Experience in developing data pipelines, evaluating AI models, and applying AI techniques to solve real-world problems. Interested in building scalable and practical AI solutions.

## SKILLS

Programming: Python, SQL, R, Java

Tools & Frameworks: TensorFlow, OpenCV, YOLOv5, Streamlit, Flask, Power BI, Tableau, Bubble.io, LangChain, Flowise, Hugging Face, Azure AI Services, Databricks

Specializations: Machine Learning, NLP, Deep Learning, Computer Vision, Generative AI, AI Agents, Data Pipelines, Model Evaluation, Data Visualization

## EXPERIENCE

### **Energys — Pune, MH**

Associate Data Science Consultant

April 2025 – Present

- Integrated Azure Bing grounding with AI agents to improve supplier discovery using real-time web data.
- Designed a supplier data pipeline to extract, structure, and score supplier information (profiles, Q&A, metadata).
- Developed evaluation methods for multi-agent workflows to improve output quality and consistency.
- Built N-tier supplier mapping to identify supplier relationships for brands and products.
- Developed an LLM-based system to extract and validate cost models from Excel data using rule-based and model-based checks.

### **AGILF(x) — Gandhinagar, GJ**

AI and ML Engineer & Certified Bubble Developer

Dec 2023 – April 2025

- Improved marketing prospecting workflows using AI for lead generation and performance tracking.
- Built a PDF chatbot for onboarding, enabling querying of internal documents.
- Developed an AI-based recruitment assistant to evaluate candidate conversations.
- Created a health chatbot for symptom analysis and nearby doctor recommendations.
- Built a regression-based model for pilot risk scoring in insurance use cases.

### **Whirlpool Asia LLP — Pune, MH**

Data Science Intern

Sept 2022 – June 2023

- Developed a YOLOv5-based model to detect dents and scratches for quality inspection.
- Automated PDF-to-Excel data extraction, improving efficiency by ~3x.

## EDUCATION

### **M.Tech in Data Science**

MIT World Peace University, Pune, MH

2021 – 2023 | CGPA: 9.60/10.0 (2nd Rank Holder)

### **B.E. in Computer Engineering**

A. D. Patel Institute of Technology, Anand, GJ

2016 – 2020 | CGPA: 8.32/10.0

## PROJECTS

- **AI-Powered Knowledge Base**  
Built a document-based chatbot using Streamlit, LangChain, and Hugging Face for querying uploaded files.
- **Research Assistant AI Agent**  
Built an AI system to automate end-to-end academic research workflows using arXiv data. Implemented a RAG-based pipeline to retrieve, process, and summarize research papers, generating structured literature reviews with insights on trends, methods, and research gaps. Enabled cross-paper querying through an interactive chatbot and supported export of reports in Markdown and JSON formats.
- **Active Learning-Based AI System**  
Developed an adaptive learning system using Streamlit and LLMs to generate quizzes from text and PDFs based on Bloom's Taxonomy. Implemented concept-level tracking, semantic answer evaluation, and an iterative learning loop ("Redemption Round") to reinforce weak areas. Included an analytics dashboard for real-time performance and mastery insights.
- **Dents & Scratches Detection (YOLOv5)**  
Trained a computer vision model using YOLOv5 and OpenCV for defect detection.
- **Indian Sign Language (ISL) Generation & Recognition**  
Developed a multi-modal system for ISL generation and validation using NLP and deep learning.
- **Handwritten Math to LaTeX Conversion**  
Built a deep learning model with a Flask interface to convert handwritten math into LaTeX.

## PUBLICATIONS

- "Indian Sign Language Generation - A Multi-modal Approach," ICCCNT, 2023.
- "A Review on Indian Sign Language Generation", 4<sup>th</sup> ICLTEM, 2023.
- "Handwritten Mathematical Equations Conversion to LaTeX," IJCTT, 2020.  
<https://ijcttjournal.org/archives/ijctt-v68i4p138>